

Dr. Waseem Iqbal
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Research Interest

Design, Synthesis of Metal-Organic Framework (MOF) and MOF-Based Mixed Matrix Membranes for Water Remediation. Heterogeneous electrocatalysts for water splitting (HER, OER, and ORR), and Electrochemical value-added productions, Metal-Organic Framework Mixed Matrix Membranes for Water remediation, and Photocatalysis, characterization of innovative polymers, membranes, and membrane electrode assemblies (MEAs) for electrolyzers and fuel cells, using both conventional and operando techniques, especially Operando Raman and Operando IR spectroscopy

Education

Ph.D. (Physical, Chemical, Material Science and Technologies)

Duration: Jan 2022 - May 2025
Board/ University: University of Calabria (<https://www.unical.it/>)
Country: Italy
Title of Ph.D. Thesis: Metal-Organic Framework Technologies for Water Remediation

Visiting Ph.D Researcher

Duration: July 2023 - Dec 2023
Board/ University: The Institute of Molecular Science (ICMol) at the University of Valencia, Valencia (<https://www.icmol.es/>)
Country: Spain

Company (Secondment)

Duration: Apr 2024 - Sep 2024
Institute/ University: Calabria Maceri e Servizi SPA (<https://www.calabramaceri.it/>)
Country: Italy

M.Phil. (Chemistry)

Duration: Mar 2016 - Feb 2018
Board/ University: Hazara University Mansehra (<https://www.hu.edu.pk/>)
Country: Pakistan
Title of MS Thesis: Synthesis, Calcination, and Characterization of Co₃O₄-Coated TiO₂ for electrical properties

BS (Chemistry)

Duration: Oct 2011 - Dec 2015
Subjects/Specialization: Chemistry
Board/ University: University of Gujrat, Gujrat (<https://uog.edu.pk/>)
Country: Pakistan

Employment Details

Postdoctoral Researcher

Employment Type: Postdoctoral Researcher
Organization: Center for Sustainable Future Technologies,
 Istituto Italiano di Tecnologia (IIT)
<https://www.iit.it/it/>

Duration: Dec 2025-Present
Job Duties: Research activity within the PROMISERS European Project focused on the development and characterization of innovative polymers, membranes, and membrane electrode assemblies (MEAs) for electrolyzers and fuel cells, using both conventional and operando techniques.

Chemistry Teacher

Employment Type: Chemistry Teacher
Organization: Mangwal Science Academy, Mangwal, Chakwal, Pakistan
Duration: Sep 2020-Jan 2022
Job Duties: During this period, I focused on teaching students at an academy in my hometown. I also pursued self-directed learning through scientific literature, online courses, and the development of research proposals in materials science and environmental chemistry.

Research Assistant

Employment Type: Research
Organization: School of Energy, Soochow University, Suzhou, China
<https://eng.suda.edu.cn/>
Duration: Sep 2018-Aug 2020
Job Duties: I was actively involved in research activities with Master's and Ph.D. students, focusing on the synthesis and characterization of heterogeneous electrocatalysts, as well as their evaluation for water splitting and the production of value-added chemicals.

Science Teacher

Employment Type: Science Teacher
Organization: Govt Primary School Sikrialia, Punjab, Pakistan
Teaching Level: Primary School students
Duration: Feb 2018 - Aug 2018
Job Duties: I was involved in teaching and supervising primary school students.

Lecturer

Employment Type: Chemistry Lecturer
Organization: Wings Group of College Chak Baili Khan, Rawalpindi, Pakistan
Teaching Level: Bachelor's degree student
Duration: Aug 2016 - Jan 2018
Job Duties: I was involved in teaching and supervising undergraduate students pursuing a bachelor's degree.

Lecturer

Employment Type: Chemistry Lecturer
Organization: Mumtaz Foundation Education Complex, Mangwal, Pakistan
Teaching Level: Intermediate Students
Duration: May 2015 - July 2016
Job Duties: I was involved in teaching and supervising Intermediate degree students.

Research Publications

1. Waseem Iqbal, Amira Nour, Rosaria Bruno, Pietro Magarò, Carmine Maletta, Rosangela Elliani,

- Antonio Tagarelli, Teresa F. Mastropietro, Jesús Ferrando-Soria, Emilio Pardo, Donatella Armentano. “Membranes Based on Metal–Organic Framework Nanostructures for Recovering Nickel, Cobalt, and Manganese Ions from Spent Lithium-Ion Batteries.” *ACS Applied Nano Materials*, 2025. <https://doi.org/10.1021/acsanm.5c04698>
2. Saad Ahmed, Christian Beauger, Amir Zada, **Waseem Iqbal**, Naveed Ahmed, Muhammad Tuoqeer Anwar, Muhammad Hassan. “Recent advancements in designing high-performance proton exchange membrane fuel cells: A comprehensive review. *Applied Energy*, 2025, vol. 390, p. 125753. <https://doi.org/10.1016/j.apenergy.2025.125753>
 3. **Waseem Iqbal**, Teresa F. Mastropietro, * Emilio Pardo, Donatella Armentano, “A Systematic Review on the Application of MOF-based Membranes Towards Wastewater Decontamination: Challenges and Opportunities” **Manuscript in preparation**
 4. Rita Maria Percoco, Rosaria Bruno, Teresa F. Mastropietro, **Waseem Iqbal**, Pegah Hajiwand, Jesús Ferrando–Soria, Emilio Pardo*, and Donatella Armentano* “In situ Studies by Single-Crystal X-ray Diffraction of Activation and isobutane Adsorption in Breathing MIL-53 (Fe) Metal–Organic Framework” **Manuscript in preparation.**
 5. Babar Shahzad, **Waseem Iqbal**, Nadia Batool, Zou Xilu, Yihui Li, Zewen Xu, Cheng Huang, Rana Muhammad Irfan, “Lithium-Rich Oxygen Batteries with Low Metal Loading Based on Iridium Single-Atom Electrocatalysts” *ACS Applied Nano Materials* 8 (15), 7463-7473; <https://doi.org/10.1021/acsanm.4c07185>
 6. Saad Ahmed, Christian Beauger, Amir Zada, **Waseem Iqbal**, Naveed Ahmed, Muhammad Tuoqeer Anwar, Muhammad Hassan, “Recent advancements in designing high-performance proton exchange membrane fuel cells: A comprehensive review”, *Applied Energy* 390(2025) 125753; <https://doi.org/10.1016/j.apenergy.2025.125753>
 7. Amira Nour, **Waseem Iqbal**, Javier Navarro-Alapont, Jesús Ferrando-Soria, Pietro Magarò, Rosangela Elliani, Antonio Tagarelli, Carmine Maletta, Teresa F Mastropietro, Emilio Pardo, Donatella Armentano, “Efficient Nickel and Cobalt Recovery by Metal–Organic Framework-Based Mixed Matrix Membranes (MMM-MOFs)”, *ACS Sustainable Chemistry & Engineering* 12 (32), 12014-12028; <https://doi.org/10.1021/acssuschemeng.4c03427>
 8. Asif Hayat, Muhammad Sohail, Sana Ben Moussa, Muhanna K Al-Muhanna, **Waseem Iqbal**, Zeeshan Ajmal, Saleem Raza, Yas Al-Hadeethi, Yasin Orooji, “State, synthesis, perspective applications, and challenges of Graphdiyne and its analogues: A review of recent research”, *Advances in Colloid and Interface Science* 319, 102969; <https://doi.org/10.1016/j.cis.2023.102969>
 9. Cristina Negro, Hector Martinez Perez-Cejuela, Ernesto Francisco Simó-Alfonso, **Waseem Iqbal**, José Manuel Herrero-Martínez, Donatella Armentano, Jesus Ferrando-Soria, Emilio Pardo, “(Multivariate)-Metal–Organic Framework for Highly Efficient Antibiotic Capture from Aquatic Environmental Matrices”, *ACS Applied Materials & Interfaces* 15 (2), 3069-3076; <https://doi.org/10.1021/acsami.2c20458>
 10. Tariq Ali, Haiyan Wang, **Waseem Iqbal**, Tariq Bashir, Rahim Shah, Yong Hu, “Electro-Synthesis of Organic Compounds with Heterogeneous Catalysis” *Advanced Science* 10 (1), 2205077; <https://doi.org/10.1002/advs.202205077>
 11. Asif Hayat, Muhammad Sohail, **Waseem Iqbal**, TA Taha, Asma M Alenad, Abdullah G Al-Sehemi, Sami Ullah, Noweir Ahmad Alghamdi, A Alhadhrami, Zeeshan Ajmal, Arkom Palamanit, WI Nawawi, Huda Salem AlSalem, Hamid Ali, Amir Zada, Mohammed A Amin, “Molecular engineering optimized carbon nitride photocatalyst for CO2 reduction to solar fuels”, *Journal of Science: Advanced Materials and Devices* 7 (4), 100483; <https://doi.org/10.1016/j.jsamd.2022.100483>
 12. Nadia Batool, Babar Shahzad, **Waseem Iqbal**, Xiao-Feng Han, Wen-Tao Wang, Jin Yan, Ruhua Shi, Jing-Hua Tian, Ruizhi Yang, “Metal–Organic Framework-Derived PtNi on N-Doped Carbon Boosting Efficient Oxygen Electrocatalysis”, *ACS Applied Energy Materials* 5 (11), 13134-

13141; <https://doi.org/10.1021/acsaem.2c02405>

13. Muhammad Sohail, Usama Anwar, TA Taha, HIA Qazi, Abdullah G Al-Sehemi, Sami Ullah, Hamed Algarni, IM Ahmed, Mohammed A Amin, Arkom Palamanit, **Waseem Iqbal**, Sarah Alharthi, WI Nawawi, Zeeshan Ajmal, Hamid Ali, Asif Hayat, “Nanostructured materials based on g-C₃N₄ for enhanced photocatalytic activity and potentials application: a review”, **Arabian Journal of Chemistry** 15 (9), 104070; <https://doi.org/10.1016/j.arabjc.2022.104070>
14. Asif Hayat, Jawad Ali Shah Syed, Abdullah G Al-Sehemi, Karam S El-Nasser, TA Taha, Ahmed A Al-Ghamdi, Mohammed A Amin, Zeeshan Ajmal, **Waseem Iqbal**, Arkom Palamanit, DI Medina, WI Nawawi, Muhammad Sohail, “State of the art advancement in rational design of g-C₃N₄ photocatalyst for efficient solar fuel transformation, environmental decontamination and future perspectives”, **International Journal of Hydrogen Energy** 47 (20), 10837-10867; <https://doi.org/10.1016/j.ijhydene.2021.11.252>
15. Shahid Khan, Tariq Ali, Xianfu Wang, **Waseem Iqbal**, Tariq Bashir, Wang Chao, He Sun, Hauliang Lu, Chenglin Yan, Rana Muhammad Irfan, “Ni₃S₂@ Ni₅P₄ nanosheets as highly productive catalyst for electrocatalytic oxygen evolution”, **Chemical Engineering Science** 247, 117020; <https://doi.org/10.1016/j.ces.2021.117020>
16. Nadia Batool, **Waseem Iqbal**, Xiao-Feng Han, Wen-Tao Wang, Hao-Tian Teng, Xiang Hao, Ruizhi Yang, Jing-Hua Tian, “Encapsulation of Pt Nanocrystals inside Pyrolyzed UiO-66-NH₂ Metal–Organic Framework Supports as Oxygen Reduction Catalysts”, **ACS Applied Nano Materials** 4 (11), 12365-12372; <https://doi.org/10.1021/acsanm.1c02829>
17. Wei Qiao, **Waseem Iqbal**, Guangming Shang, Dan Wang, Yongwang Li, Flemming Besenbacher, Hans Niemantsverdriet, Chenglin Yan, Ren Su, “Paired Electrochemical N–N Coupling Employing a Surface-Hydroxylated Ni₃Fe-MOF-OH Bifunctional Electrocatalyst with Enhanced Adsorption of Nitroarenes”, **ACS Catalysis** 11 (21), 13510-13518; <https://doi.org/10.1021/acscatal.1c03938>

Conference & Summer Schools, and Workshops Attended

1. **Summary School in 34th Meeting of the European Crystallographic Association (ECM 34)**
Your role: Poster Presentation
City/Country: Padova, Italy
Conference Date: Aug 26-30, 2024
<https://www.ecm34.org/abstracts>
2. **1st Mediterranean Conference on Porous Materials, MEDPore 23**
Your role: Poster Presentation
City/Country: Crete, Greece
Conference Date: May 17-19, 2023
<https://medpore.eventsadmin.com/Home/Welcome>
3. **Virtual International Workshop on Energy Storage Technologies for E-Mobility (IESETE-2021)**
Your role: Participation
City/Country: Department of Chemistry, SRMIST, India
Conference Date: Mar 25-27, 2021
4. **Wiley-Soochow University international conference on Energy Materials**
Your role: Participation
City/Country: Suzhou Institute of Materials and Energy Innovations, China
Conference Date: Dec 21-23, 2018
<https://imee.gdu.edu.cn/info/1192/1903.htm>
5. **Two-Day Workshop on Nanomaterials Research, Fabrications and Applications**

Your role: Participation
City/Country: Department of Chemistry, Hazara University, Mansehra, Pakistan
Conference Date: Oct 12-13, 2017

6. First National Conference on Emerging Trends in Materials Science

Your role: Participation
City/Country: Abbottabad University, Pakistan
Conference Date: Oct 05-07, 2017

Skills

Material Synthesis Methods: Hydrothermal, Co-precipitation, Sol-Gel, Solvothermal, Microwaves

Material Characterization: SEM, XRD, FTIR, RAMAN, Mechanical characterization of Membranes and Electrochemical Workstation

Software Skills:

Microsoft Office, PowerPoint, MS Excel, Adobe (Adobe Photoshop and Adobe Illustrator), Chem-Draw, Origin, EndNote, Jade 6.5, ImageJ, Mercury, X'Pert High Score Plus, Crystal Maker, Blender

Communication Skills:

English, Urdu, Punjabi, Basic Chinese, Basic Italian, and Basic French

Competition Participated

1. Selected for Poster Presentation in 2024 French MOFs, COFs and Porous Polymers Conference

City/Country: Lyon Tech La Doua Campus, Villeurbanne, France
Date: Jun10-11, 2024
<https://frenchmof2024.sciencesconf.org/>

2. Selected for Poster Presentation in Unical Ph.D. Day

City/Country: University of Calabria, Italy
Date: May 30, 2023
<https://www.unical.it/didattica/offerta-formativa/dottorati/phd-day/>

3. 2nd position from essay writing competition in district Chakwal

City/Country: Department of Education, Punjab, Pakistan
Date: 2009

Awards, Scholarships, and Fund Achievements

1. Scholarship Award

Type of Award: Scholarship
Awarding Body: 37th Cycle, Italian National Scholarship (PON), European Union Project
Supporting Program: Ph.D.
Duration of Award January 2022- December 2024

2. Prime Minister's Pakistan laptop scheme for talented students.

Type of Award: Awarded by Laptop
Awarding Body: Government of Pakistan
Supporting Program: M.Phil. Chemistry
Awarding Year 2018

3. Chief Minister of Punjab's laptop scheme for talented students.

Type of Award: Awarded by Laptop
Awarding Body: Provisional Government of Punjab, Pakistan
Supporting Program: BS Chemistry
Awarding Year 2012

References

Dr. Donatella Armentano

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Dr. Teresa Fina Mastropietro

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Dr. Wajid Rehman

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